Project Progress Report

MedXVLM — explained in plain English

B.Sc. AI & Data Science — Major Project

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What is this project, in one paragraph?

Doctors write a report every time they look at a chest X-ray, describing what they see (or don't see). This project is trying to build an AI system that can look at a chest X-ray and do three things: (1) tell if the picture and the report actually "match" each other, (2) guess which of 14 common problems (like an enlarged heart, fluid in the lungs, a collapsed lung, etc.) are visible in the image, and (3) eventually write a short report of its own, in plain English, describing what it sees. On top of all that, the system is also meant to be able to show *why* it made a decision — for example, by highlighting the part of the image it was actually looking at — instead of just spitting out an answer with no explanation.

It's built in three stages, one on top of the other, like building blocks:

Stage	What it does
Stage 1	Teaches the AI to connect an X-ray image with its written report — i.e. learn what "looks like" what "reads like".
Stage 2	Uses what Stage 1 learned to guess which of the 14 known conditions are present in a new X-ray.
Stage 3	Uses the same understanding to write a short report describing the X-ray in words.

Every stage also gets an "explain yourself" layer added on, so a doctor (or a grader) can check whether the AI is actually looking at the right part of the lungs, instead of just trusting a number.

Where things stand right now

> So far, I've finished the planning and the "skeleton" of the code. Nothing has been trained on real data yet, and no real predictions have been

> made yet — that comes next. What exists right now is proof that all the pieces fit together correctly and can run without crashing.

Phase 1 — Getting ready DONE

Before writing any AI code, I spent this phase making sure the groundwork was solid:

- **Wrote a project proposal** explaining what the project is, why it matters, and what I'm hoping to find out from it.
- **Read background papers** on chest X-ray AI, so my design choices (like which base model to start from) are backed by what's already known to work, not just guesses.
- **Picked a dataset** — a public, freely available collection of about 7,470 chest X-rays with their real written reports (called the Indiana University / Open-i dataset), and downloaded it.
- **Checked my computer and software actually work** — made sure my graphics card (GPU), which is what makes training AI models fast, is properly detected and every required software library installs and runs correctly. This sounds trivial but is a very common place for projects to quietly break later if skipped.
- **Wrote a tiny test script** that just opens a few images and their matching reports side by side, to prove the raw downloaded data can actually be read by code at all.

Phase 2 — Building the model "skeletons" DONE

This phase was about building the actual building blocks of the AI model — but only enough of each one to prove they're built correctly, not enough to actually be good at the task yet. Think of it like building the frame and wiring of a house before you've painted anything or moved in furniture: you want to know the frame doesn't collapse and the electricity works, before spending time on how it looks.

Concretely, I built:

- **A "before" comparison model** — a simple, plain image classifier with no fancy tricks, representing how people used to do this a few

years ago. This exists so that later, when my real model is trained, I can prove it's actually better than the old-fashioned approach and not just different.

- **An image-understanding module** — takes a chest X-ray and turns it into a set of numbers the AI can reason about, while also keeping track of *which part* of the image each set of numbers came from (so later, the AI can point at a specific region of the lung, not just say "yes/no" for the whole picture).
- **A text-understanding module** — does the same thing, but for the written report instead of the image, turning sentences into numbers the AI can compare against the image numbers.
- **A "matching" module (Stage 1)** — takes the image numbers and the text numbers and trains them to end up close together when they belong to the same X-ray/report pair, and far apart otherwise. This is the actual "learning to connect pictures and words" part.
- **A "diagnosis" module (Stage 2)** — takes the image numbers and, for each of the 14 possible conditions, asks "does this specific condition seem present?", giving 14 separate yes/no-style scores.
- **A test runner** that feeds a handful of real X-ray images and reports through all of the above, on purpose in a "trying to memorize just this one small batch" way, and checks three things: nothing produces broken numbers (like NaN, which is a common silent-failure sign in AI code), every part of the model that's supposed to be learning is actually receiving feedback (nothing is silently disconnected and doing nothing), and the model's error score actually goes down over repeated attempts — proof that learning is genuinely possible through every wire in the system, not just that the code runs without crashing.

> **Result:** I ran this test and it caught a real bug on the first try — part of my text-understanding module had a small piece that was built but never actually used, meaning it would never have learned anything even during real training. I found and fixed it before it could waste an entire training run down the line. This is exactly the kind of problem this test was built to catch, so I'd count that as the test doing its job.

What's deliberately *not* done yet LATER

A few things are intentionally left for later stages, so it's worth being upfront about them instead of implying more progress than there is:

> - No real training has happened. Everything above was tested with either a handful of real images or made-up random data, run for a tiny number of steps — enough to prove the wiring works, not enough to produce a usable model.
> - The AI doesn't have real labels to learn from yet for the 14 conditions — that requires building a proper labeling tool that reads each report and decides which conditions it mentions, which is the very next thing on my list.
> - Stage 3 (writing reports in its own words) hasn't been started at all.
> - None of the "explain yourself" / highlighting features have been built yet — they depend on Stages 1 and 2 actually being trained first.
> - No accuracy numbers exist yet, because there's nothing trained to measure accuracy on. Any percentage I might mention right now would be meaningless.

What's coming next

Next up	In plain terms
Build the real labeling tool	Write code that reads each report and decides, for each of the 14 conditions, whether it's mentioned as present, absent, or unclear.
Split the data properly	Divide the X-rays into a "training" set (to learn from) and a "test" set (to check honestly, on X-rays it's never seen) — making sure the same patient never appears in both, which would be a form of cheating.
Actually train Stage 1 and Stage 2	Run the real training process on the real data for a proper amount of time, using the building blocks from Phase 2.

Measure how good it is	Compare the trained model's accuracy against the simple "before" model built in Phase 2, to see if the fancier approach was actually worth it.
Add the explain-yourself features	Generate heatmaps showing which part of the X-ray the model focused on for each decision.
Build Stage 3	Get the model writing its own short reports, and check if they make clinical sense.

In short

I've spent the first two phases making sure the foundation is solid: a justified plan, a working dataset, a working environment, and a carefully tested skeleton of the actual AI architecture, with the wiring between all the pieces verified to actually work rather than just assumed to work. Nothing has been trained for real yet, and no results exist yet — that honestly is the point of this stage. The next phase is where real training, real numbers, and real evaluation start.